University of South Wales

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ABSTRACT

The continuous growth in flight operations has led to public concern regarding the impact of aviation emission on climate change. There is also a need for technological advancements in extending drone technology to places where humans can't reach such as calamity struck areas and to help deliver medicines in rural regions. Combining these two ideas have propelled to simulate a drone model and battery modelling. Although several papers are limited to either of the models, this paper aims to develop a drone model capable of electric propulsion and a battery model to power the electric need. By providing the concept of integrating two novel ideas, lay a platform to a new research potential of further studies. The proposed drone model is made modular to be applicable for an aircraft and the battery model to propel any carrier by increasing the battery packs. Also, it is necessary to identify the individual contribution from each power source and optimize each system for the overall performance. The designed power management strategy will effectively direct the power flow between power sources to satisfy the load demand in each flight phase. The novelty to the study comes by combining all three models and letting them operate co-dependently whose modularity also contributes to its singularity. The integrated system model will be used to perform several static and dynamic ground tests both on simulation later confirmed by experiment. The various system level test data will be collected and analyzed. Also, the collected experimental test results will be compared with the simulation results to provide further suggestions and improvements to power distribution strategy and overall propulsion system.

Keywords: Computational Fluid Dynamics, Gas Turbine, Bell mouth design and modelling, ANSYS fluent.

1. INTRODUCTION

Main aim of this research is to find out an efficient design that can increase the aerodynamic performance of an aircraft engine. This means that there should be minimum losses in pressure. Minimum pressure loss can be achieved in many ways like guided airflow such that there is no loss of energy. One of the ways of perfectly guiding the inlet airflow is to use a bell mouth at the mouth of the aircraft engine. This bell mouth will suck in more air for the inlet and guides it in a certain direction so that there is the maximum amount of air being ingested at the inlet thus increasing the aerodynamic efficiency of the aircraft engine. Initially, a bell mouth structure is designed using design software and imported into ANSYS fluent for further computational fluid dynamics analysis. The model is then computationally analysed using ANSYS fluent and the results of the simulation/analysis are compared with the baseline data, that is, the aerodynamic performance of an aircraft engine without a bell mouth. There are many different types of bell mouth designs. This research also considers the existing bell mouth design, and a unique and efficient bell mouth design is modeled and adopted for the purpose of this research. From CFD analysis conducted it was concluded that the bell mouth attached at the inlet of the engine minimises the pressure loss and hence increases the aerodynamic efficiency of the engine.

The gas turbine engine is one of the most efficient forms of generating energy in the form of lift and thrust. It has been adapted by the aircraft industry in various forms such as turbojet, turboprop and turbofan engines based on their application at different air velocities, altitudes, and efficiencies. For an idealized gas turbine engine, the operation is based on Bryton cycle where atmospheric air is used as the working fluid. The engine converts the energy of high-speed air and fuel to mechanical energy in the form of thrust and lift. The inlet air is first compressed to high pressure at a compressor stage. The pressure of air is increased while the flow velocity is maintained constant. The air then enters the combustion chamber where ignition takes place. The fuel is injected through jet injection nozzles along with fresh cool air through the fuel liner. The air-fuel mixture is then ignited to generate combustion. This is followed by an expansion phase where the energy of gas is extracted in the form of mechanical work using a turbine. The hot gas is passed through airfoil shaped blades along multiple stages. The blades convert the energy of gases into mechanical energy by rotating a shaft. The gas exits the turbine through a nozzle which acts as the exhaust phase.

There are several design challenges which must be addressed in order improve the efficiency of the turbine. Improvements can be made at each stage to optimize the energy output of the turbine. For example, adding a bell mouth at the inlet to increase the volume of air entering the engine. It consists of a convergent section which terminates at the compressor inlet. This can be particularly helpful at higher altitudes where the air pressure is low. Similarly, a multistage compressor with a low-pressure phase and a high-pressure phase can be used to increase the air pressure entering the turbine. Although this would require more energy input, the increased air pressure could be beneficial for combustion. Further, applying the recent advancements in combustion technology, design modifications can be made to the combustion chamber to facilitate complete combustion, thus increasing the overall energy output from the turbine. Similarly, a multistage turbine can be incorporated to extract more energy from the air. The compressor and turbine outputs can further be increased by incorporating intercooling and reheating. Fresh cold air is introduced to the compressor after the low-pressure stage. This reduces the air temperature which makes it possible to be compressed to a higher pressure at the high-pressure stage. Similarly hot flue gases after combustion is introduced after the high pressure stage of the turbine. This heats up the air, increasing its internal energy. Hence, more work can be extracted by the low-pressure turbine thus resulting in an overall increase in efficiency. Afterburners can be employed to further inject trace amounts of fuel at the exhaust phase to maximize the energy extracted from the air leaving the turbine. Each improvement has its own set of design goals to optimize over a set of design variables with the overall objective of maximizing energy extracted from the system

In this dissertation, the bell mouth design problem is studied for a gas turbine engine. Adding a bell mouth increases the mass flow rate of air at the inlet of the gas turbine engine without any increase in the pressure. Further, due to its smooth gradual inlet section, the turbulence of air at the compressor inlet is greatly reduced. Hence the bell mouth cross section and length of the bell mouth inlet section and some of the major design parameters to optimize the gas turbine engine performance. In this study, the bell mouth design is optimized as a stand-alone component and the effects of adding a bell mouth on the downstream components is also analyzed. Various design parameters such as the bell mouth inlet length, taper angle, geometry of the cross section etc. are studied to determine the best bell mouth geometry which maximizes the engine performance.

Bell mouth designs:

There are three main types of bell mouth designs. Any bell mouth design can be easily classified into one of the three below categories.

- Radius profile
- Airfoil profile
- Elliptical profile

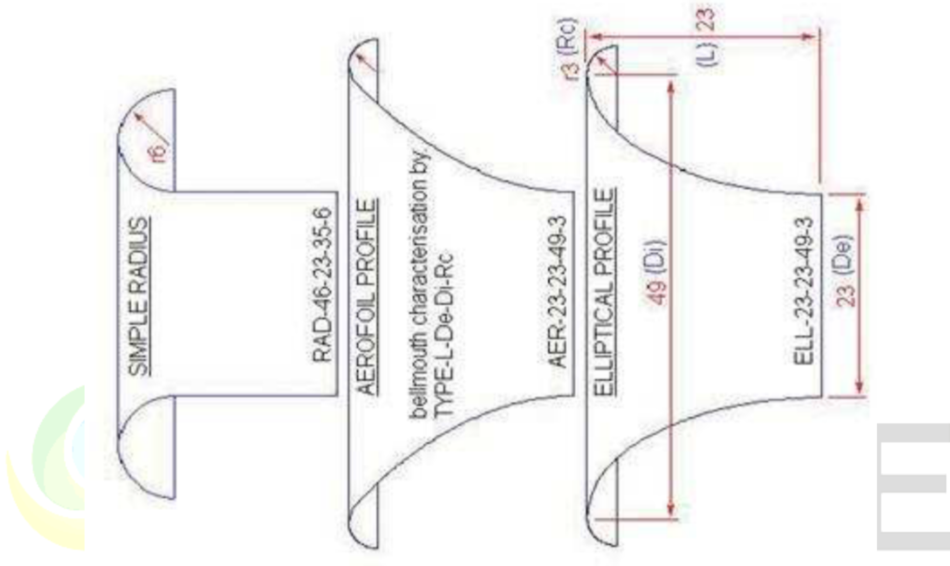

Figure 1: Different types of bell mouth designs

An efficient bell mouth design for the purpose of increasing the aerodynamic efficiency is decided by calculating the coefficient of discharge of all the three mentioned bell mouth profiles. After further literature review and data collection from research it was found that the elliptical profile bell mouth design was the best for the purpose of increasing the aerodynamic efficiency of the aircraft engine.

The bell mouth geometry is designed with a uniform radius (bell mouth radius) for air inlet with an outward flare (arc radius) to reduce losses at the beginning of the air inlet. The bell mouth is attached to the inlet guide vanes (inlet radius) at the tangent to ensure a smooth connection as shown in Figure 1.

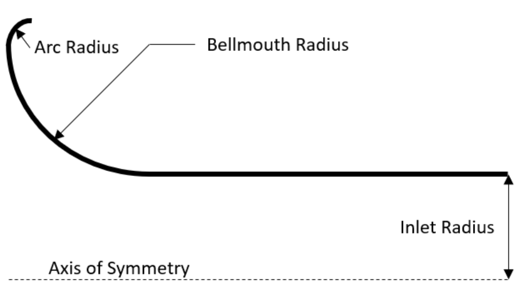

Figure 2: Bellmouth geometry

An optimization problem can be set up based on selected geometric design parameters to maximize the engine performance such as thrust and lift. The dimensions of the design parameters used to generate a baseline geometry is listed is Table 1.

Design details of the bell mouth:

Inlet diameter$D_i = 322.2\ mm$

Exit diameter $D_e = 150\ mm$

Generally, exit diameter D_e is 2.14 times smaller than the inlet diameter.

Mass flow rate = 1kg/s

Density of air = 1.225 kg/m^3

Pressure of air = 101325 Pa

$$v_e = \frac{\dot{m}}{\rho Ae}$$

Using Bernoulli's equation: $P_1 + \frac{1}{2}\rho {v_1}^2 = P_2 + \frac{1}{2}\rho {v_2}^2$ = constant

2. ANALYSIS SETUP

This section talks about a detailed step by step process of how to perform the analysis in a computational fluid dynamics analysis tool, namely, ANSYS Fluent, and obtain the results. In this part, the environment is set up to perform the computational fluid dynamic analysis.

Firstly, the bell mouth structure is designed in solidworks. The design details are as given below:
$D_i =$
$D_o =$
$r =$
$L =$
$l_1 =$
$l_2 =$

All dimensions are in mm

The cross section of the bell mouth is taken to be circular geometry of diameter of 0.45 m at the inlet. The length of the inlet section of the bell mouth is taken to be 0.09 m with a gradual slope defined by a circle of radius 0.09m as shown in Figure 1. The compressor inlet is attached at the end of this taper region. The length of the engine including the compressor section, combustion chamber and turbine section is taken as 1.1 m. The exhaust (after burner section) has a taper length of 0.295 m with a circular outlet diameter of 0.1 m. The moving components of the engine such as the compressor and turbine blades, fuel injection etc. are assumed to have no significant effect on the bell mouth design and hence are not included in this geometry. This makes the finite element analysis more accurate for the bell mouth study. Hence the effect of adding a bell mouth at the compressor inlet can be abstracted through the two-dimensional design shown in Figure 1 and the computational design and analysis process outlined in this study is still valid for the bell mouth. The design was realized using ANSYS design modeler and exported for meshing.

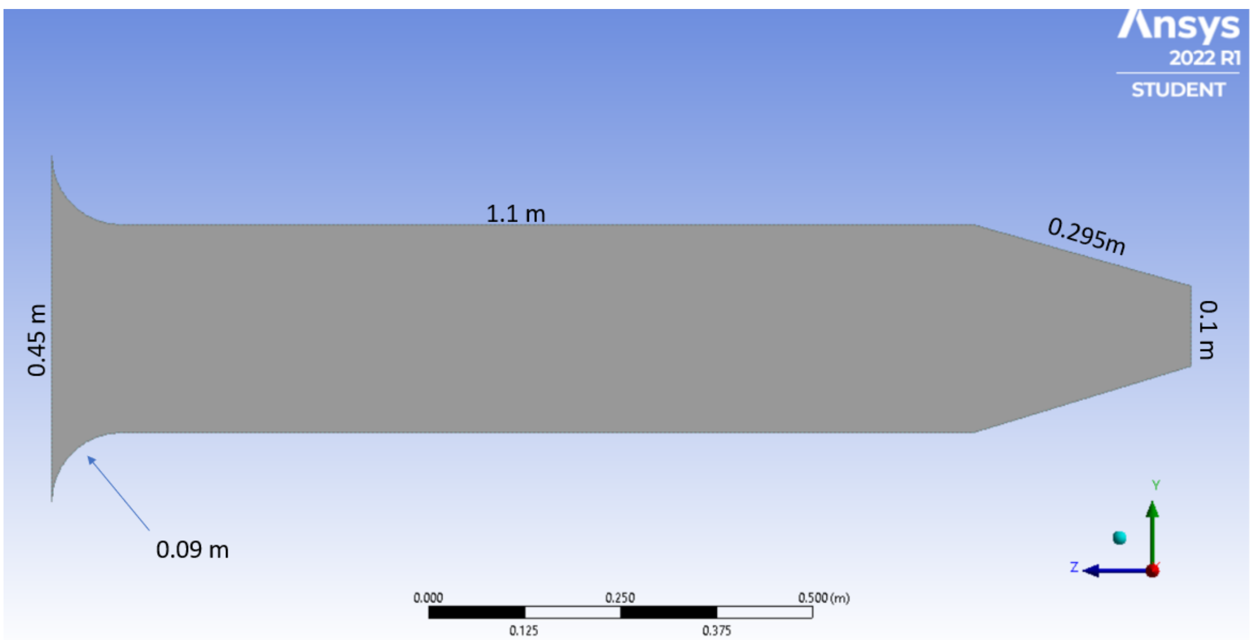

Figure 3 Initial geometry of the engine fitted with a bell mouth of circular cross section at the inlet (inlet diameter of 0.45 m) with a gradual taper angle (defined by a circle of radius 0.09 m)

This design of the bell mouth forms the geometry of the computational fluid dynamic analysis done using ANSYS fluent. The design for the selected bell mouth is done in ANSYS fluent design modeler as it is a fairly simple design. The final design is verified such that it does not have any missing lines or sections. Once the correctness of the design is confirmed then the next step was to create the 'Named section'. Named sections are where all the sections of the geometry are given a proper label. It is very important to create this named section, that is, a labelled section, which will be the ones to be used to establish boundary conditions around the bell mouth. The named sections created for the selected bell mouth design are as given below.

To create these named sections, we have to first select the edge selection command from the toolbox. After that, each edge is selected, and the desired label is given to that particular edge. In this case, the upper wall of the bell mouth, aircraft engine, and the nozzle is selected, and all these edges are marked as an upper wall. Similarly, the lower wall of the bell mouth, aircraft engine, and the nozzle is selected, and these edges are marked as a lower wall. The entire inlet edge of the bell mouth is selected and labelled as inlet and the outlet edge of the nozzle is selected and labelled as an outlet. The inner portion of the bell mouth, aircraft engine, and the nozzle are by default labelled as the interior surface body. These named sections are important when boundary conditions are being defined in the set up section of the numerical analysis.

Once the geometry is created and the named sections are defined accurately, we proceed to the next step of the numerical analysis, that is, meshing. When the meshing tab is opened, it automatically imports the geometry created along with all the named sections.

Meshing

In finite element analysis, the domain of interest (the engine with bell mouth geometry in this study), is discretized into several discrete elements which are representative of the bulk. The elements are defined by its nodes and geometry. For a triangular mesh, the elements have three nodes with a triangular geometry. Further, the elements can be classified by their dimension. For a two-dimensional analysis, the discretization element should also be two dimensional with similar properties. Further, the elements are to be selected based on their behavior in order to closely resemble the behavior of the overall system. If the problem of interest in a hollow beam, a solid element will not capture the physics accurately. Hence, selection of the right mesh element is an important process in finite element analysis. The number of discretization elements plays an important role in the accuracy of the results. For complex geometries, the higher the number of elements, the more accurate the results with be. This also increases the computational complexity of the problem. Hence, the tradeoff between accuracy and computational power needs to be accounted for. To address this tradeoff, several design tools can be employed such as mesh refinement, inflation layers etc. Refinement adds more elements near the edges of geometries and in locations where there is a sudden change in the geometry while the number of elements in the bulk of the geometry is not modified. Adding inflation layers near the walls can help capture the boundary layer phenomenon only near the walls without having to increase the elements on the bulk of the geometry. Mesh metrics such as orthogonality and skewness can be directly related the quality of the mesh for a given geometry. Using such metrics, the mesh parameters can be tuned to get the optimized results. Orthogonal quality refers to how close the angle between each element is. Ideally, the angle between elements should be zero which relates to continuity of the mesh. High orthogonal angle leads to contact separation issues and end up with large inaccuracies between the finite element analysis and observed physics. Similarly, a skewness value of zero is ideal since it corresponds to each mesh element being equilateral to each other. A target skewness very close to zero would satisfy all the assumptions made by the solver when performing the finite element analysis. Ideally, mesh elements should have a target skewness under 0.1 with an orthogonal quality of over 90%.

For discretization of the engine geometry with the bell mouth, an unstructured triangular mesh was adopted with element size of 10 mm. Figure 2 shows the mesh structure adopted for this design. Using mesh metrics, the quality of the mesh is analyzed. For a free unstructured mesh using triangular elements, the orthogonality and skewness conditions are taken as the major parameters and the properties of the mesh such as mesh size, refinement, inflation layers and geometry of the mesh element are selected to optimize for these metrics. The resulting mesh was composed of 4085 nodes and 7805 elements with a target skewness of under 0.1 and an orthogonal quality of more than 95% as shown in Figure 3 and 4 respectively.

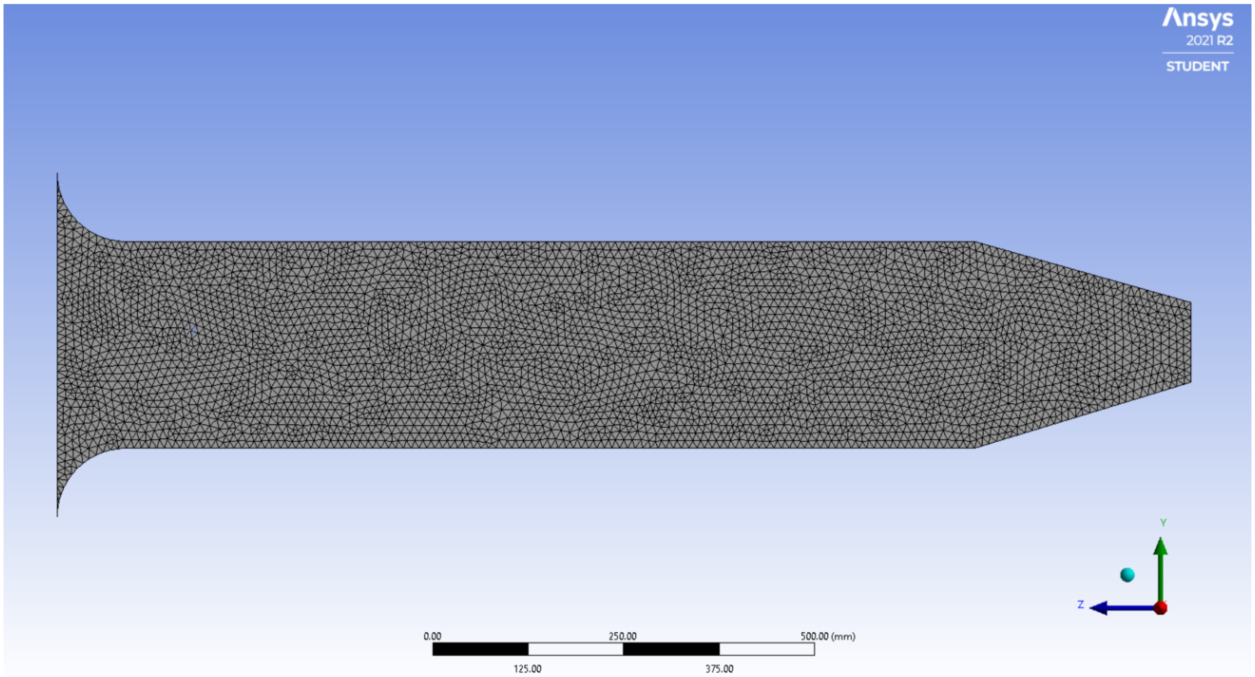

Figure 4: Discretization of the gas turbine geometry with a bell mouth

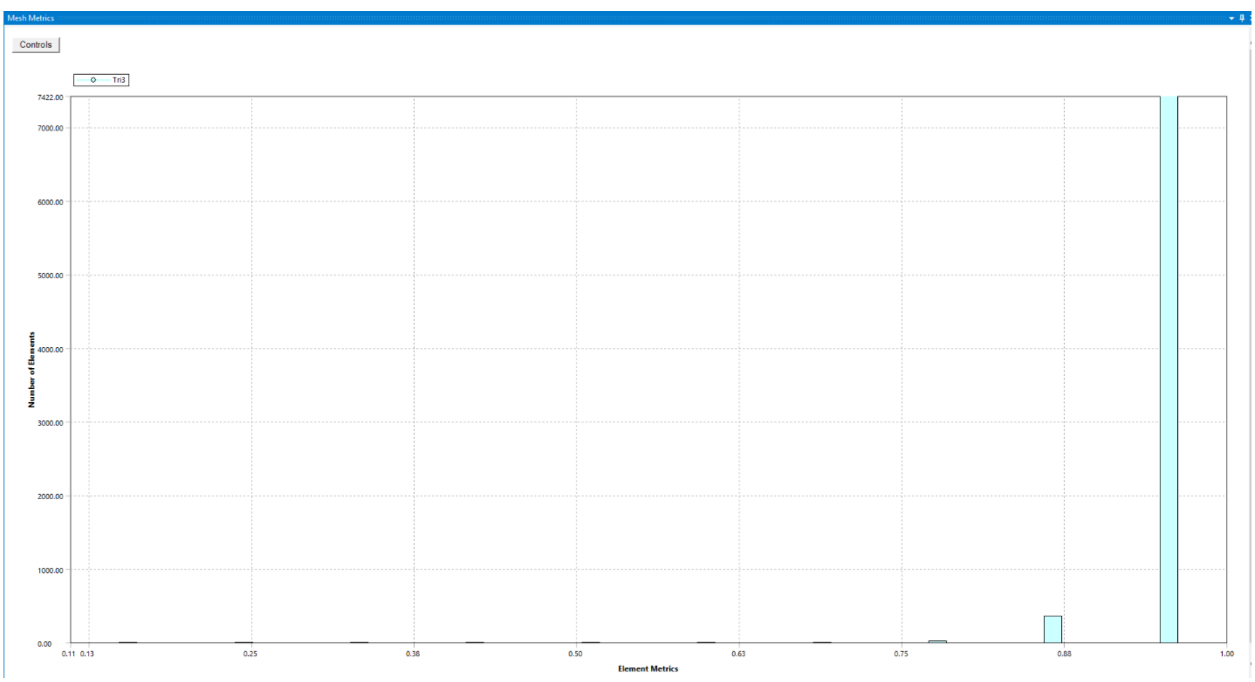

Figure 5: Orthogonal quality of more than 95% for the discretization method

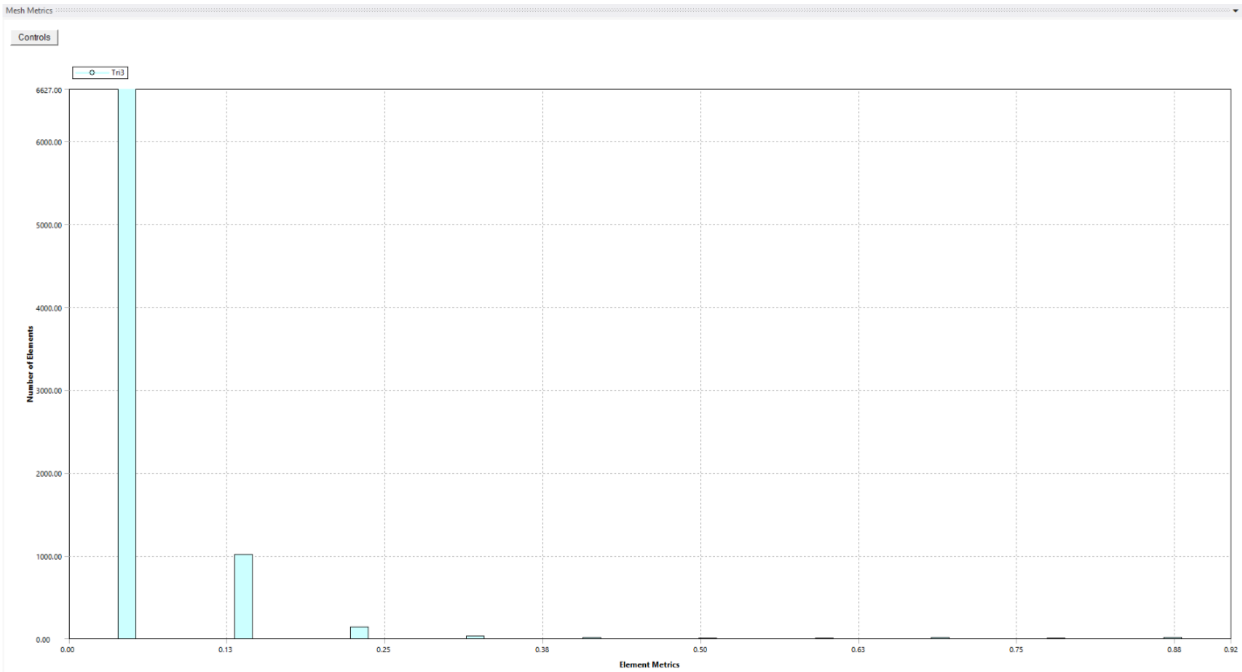

Figure 6: Target skewness of less than 0.1 for the discretization method

A computational fluid dynamical model has to be selected such that the analysis performed accurately imitates the real world process. In this case, a viscous SKE model is being used.

The fluid surrounding the model is set to be air and the properties of air are set as given below:

- Density (ρ) is set to be 1.2256 Kg/m^3
- Pressure (P) is set to be 101325 Pa
- Mass flow rate ($\dot{m}$) is set to be 1 Kg/s

After the fluid around the body is selected and its properties are set accordingly the next step is to set and/or define the boundary conditions for the model. Setting these boundary conditions properly is very important as these are the attributes that define how the body is going to behave when subjected to fluid flow around them. These boundary conditions will make sure to replicate the behaviour of the fluid (in this case, air) around the body in terms of numerical analysis. This is where the named section created in the geometry section becomes useful. The created named sections will be the ones that will be used as reference names to define the boundary conditions. In this analysis, the boundary conditions for the bell mouth attached to the aircraft engine are set as given below:

- Inlet: The inlet is set to type, pressure inlet. The reference frame is set relative to the adjacent cell zone, gauge total pressure is set to 101325 Pa, initial gauge pressure is set to 101325 Pa. Direction is specified to be normal to the boundary. Under the turbulence section, intensity and viscosity ratio is chosen as the specification method used, turbulent intensity is set to 5%, turbulent viscosity ratio is set to 10. Under the turbulence section, the specification method I set to be intensity and viscosity ratio.

- Outlet: Outlet section of the bell mouth is set as a pressure outlet where it represents all the space around it as pressure of the air right when the air is exiting the engine. This pressure outlet will have to be set as the outlet pressure. This pressure outlet is supposed to be calculated for the same is given below using Bernoulli's equation:

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 =$ constant

v_1 = 0 (assumed)
v_2 = 90 m/s (considered)
P_1 = 1.2256 Kg/m^3

Therefor, $101325 + 0 = P_2$ $+0.5 * 1.225 * 90^2$

The pressure ratio is assumed to lie in the range of 1 to 1.1 for optimal results, that is, the aircraft has optimal or efficient aerodynamic performance at this particular pressure ratio.

- Upper wall: Upper wall is the section which is set to be type wall. Here wall is set as stationary that is zero velocity and has no slip condition. Under the roughness models, roughness height is set to zero and roughness constant is set to 0.5

- Lower wall: The lower wall is the section that is set to be type wall. Here the wall is set to be stationary, that is, zero velocity with no shear condition. Similar to the upper wall set up, roughness height is set to zero, and the roughness constant is set to 0.5

- Internal body surface: Interior section of the body is set as interior. Internal surface of the body is a section that is defined by the chosen fluid, in this case, the chosen fluid type is air with the properties defined earlier.

All the other values from heat flux, roughness height, etc., are set to zero.
Once the boundary conditions are set next step is 'Solution Initialization'. Here the computational fluid dynamic analysis is set to be computed from 'inlet' which is set to be velocity inlet with the fluid or air velocity set to 90 m/s.

After this the 'Reference frame' is set to 'Relative to Cell Zone'. The initialisation values are set automatically. Once these values are set properly, then there are two options namely, hybrid initialisation and standard initialisation. Standard initialisation fills the values of fluid properties and makes sure that these values are fixed and do not change. On the other hand hybrid initialisation will start with the concept of standard initialisation where the fixed values are for the field properties and then these values are updated where it solves a set (ten iterations) number of iterations for a simplified system. The data obtained from initialisation gets built on in the 'calculation' phase. This process will make sure that the calculated value at the end of each iteration is better (with least errors) compared to standard initialisation. In this case hybrid initialisation is chosen as it gives better and comparatively more accurate results. Once these values are decided and set then in the iteration window the iteration rate is around two seconds for each and every iteration. In this analysis the number of iterations is set to be 1000 which takes about an hour to hour and a half depending on the computation power or capacity of the computer. The number of iterations can be changed either by increasing it or decreasing it. But the main aim is to make sure that all the values which are being analysed reach convergence. These values are said to have reached convergence when the value of error between the previous iteration value and the current iteration value is less than the set error limit (1×10^{-6}). These values do not converge unless the error value is less than or equal to 1×10^{-6}.

2.1. Solver method used

SKE model gives better results for velocity profile (Jehad et al, 2015). Since in this study we aim to compare with the velocity profiles obtained from engine only numerical analysis and bell mouth intake attached to the engine analysis. Hence SKE model is preferred for this particular analysis.

Transport equation for SKE model:

For turbulent kinetic energy k

$$\frac{\partial k}{\partial t} + \frac{\partial (k u_i)}{\partial t} = \frac{\partial}{\partial x_i}\left[\left(\frac{\mu}{\rho} + \frac{\mu_t}{\rho \sigma}\right)\frac{\partial k}{\partial x_i}\right] + P_k + P_b - \varepsilon - Y_m + S_k$$

Approximating to the problem

$$u_i \frac{\partial k}{\partial x_i} - \frac{\partial}{\partial x_i}\left[\frac{\nu_T}{\sigma_k}\frac{\partial k}{\partial x_i}\right] = P_k - \varepsilon$$

For dissipation ε

$$\frac{\partial (\rho \varepsilon)}{\partial t} + \frac{\partial}{\partial x_i}(\rho \varepsilon u_j) = \frac{\partial}{\partial x_i}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_i}\right] + C_{1\varepsilon}\frac{\varepsilon}{k}(P_k + C_{3\varepsilon}P_b) - C_{2\varepsilon}\left(\rho \frac{\varepsilon^2}{k}\right) + S_\varepsilon$$

Approximating to the problem

$$\left(\div \rho, P_b = 0,\ S_\varepsilon = 0, \frac{\partial \rho}{\partial t} = 0, \frac{\partial \mu}{\partial x_i} = 0\ \right)$$

$$u_j \frac{\partial \varepsilon}{\partial x_i} - \frac{\partial}{\partial x_i}\left[\frac{\nu_T}{\sigma_\epsilon}\frac{\partial \varepsilon}{\partial x_i}\right] = \frac{\varepsilon}{k}(C_{1\varepsilon}P - C_{2\varepsilon}\epsilon) \tag{14}$$

$C_\mu = 0.09$; $C_{1\varepsilon} = 1.44$; $C_{2\varepsilon} = 1.92$;$\sigma_k = 1$;$\sigma_\epsilon = 1.3$

2.2. Step by step description of analysis set up

Once the analysis and its calculation are complete, it is important to check if convergence has been reached or not. If convergence has not been reached, then the same calculation has to be repeated for a greater number of iterations until convergence is reached (or can be reached within reason). Once convergence is reached then the results obtained from this computational analysis can be visualised in the results section of fluent. In the results section there are different ways to realise and visualise the results of the numerical analysis.

To visualise how pressure distribution looks like after the numerical analysis. For this the contour option is used and static pressure option is used. The static pressure distribution around the internal surface. This gives the static pressure, dynamic pressure distribution around the body. Pressure contour lines are also known as isobars. These are the lines with same pressure or constant pressure. More about how the static pressure and dynamic pressure distribution looks like will be explained in the results and discussion subsection of Analysis section. From Bernoulli's equation given by

$P + \frac{1}{2}\rho v^2$ = constant

It is evident that pressure is inversely proportional to the velocity in a given region. Similarly, contour options can also be used to visualise the temperature distribution. These contours are colour coded. Any region on the body with the same colour indicated the contour lines called as isotherms. These are the lines with constant and same temperature. These isotherms are vital to be indicated as they are constant and not changing, they give a perfect representation of constant temperature distribution about the body along with the indication of which region has maximum temperature and which one is with minimum temperature.

Velocity contour

Velocity is the most important factor that has the potential to justify the selected design parameters for the bell mouth intake. We have already established that Bernoulli's principle that total pressure at any point chosen inside the internal surface body will remain a constant. For a velocity contour, contour option under graphics is selected. In the contours dialog box, under contours of heading, velocity is selected and then there are many other options like velocity magnitude, x velocity, y velocity, radial velocity, relative velocity, etc., Amongst this velocity magnitude is selected and this has to be applied onto the interior surface body and the contour is completed. Similarly velocity contour of x velocity, stream flow and relative x velocity is computed, and the contour is saved and displayed. The significance of this and its interpretation is explained in detail in the results and discussion subsection under analysis section.